

where $\mu_t^{1 \rightarrow 2}$ is as in (11) and

$$\mathcal{W}_\nu^2(\mu_1, \mu_2) := \int_{\mathbb{R}^d} \|T_\nu^1(x) - T_\nu^2(x)\|^2 d\nu(x) \geq \mathcal{W}_2^2(\mu_1, \mu_2). \quad (13)$$

Note that the definition implies the following property which is useful in our analysis, $\forall t \in [0, 1]$,

$$\phi(\mu_t^{1 \rightarrow 2}) \leq (1-t)\phi(\mu_1) + t\phi(\mu_2) - \frac{\lambda}{2}t(1-t)\mathcal{W}_2^2(\mu_1, \mu_2). \quad (14)$$

The definition of convexity a.g.g. in [7, Section 9.2] is for the more general case when ν may not have density and the OT maps from ν to μ_i need to be replaced with optimal plans, and then the generalized geodesics may not be unique. In this paper, we only consider the case where ν has a density, so we simplify the definition, see [7, Remark 9.2.3] (and make it slightly weaker, but there is no harm for our purpose).

We also note that many functionals $\phi(\mu)$ on $\mathcal{P}_2$ that are geodesically convex actually also satisfy the convexity a.g.g. in Definition 2.3. Examples include $\phi(\mu) = \int V(x)d\mu(x)$ with V convex on $\mathbb{R}^d$, $\phi(\mu) = \iint W(x_1, x_2)d\mu(x_1)d\mu(x_2)$ with convex W , and $\phi(\rho) = \int F(\rho(x))dx$, ρ being the density, where F is convex on $[0, \infty)$. (In these examples, V , W , and F need to satisfy other technical conditions.) The last example includes negative entropy as a special case, where $F(x) = x \log x$. The primary case for our work is when ϕ is the KL divergence, which will be discussed in more detail in Section 4.1. We refer to [7, Section 9] for other examples and detailed discussions of convex a.g.g. functionals.

2.3 JKO scheme for Fokker-Planck equations

Consider the diffusion process (7) starting from $P \in \mathcal{P}_2$. It is known that under generic conditions, as $t \rightarrow \infty$, ρ_t converges to the equilibrium distribution of (7) which has density

$$q \propto e^{-V}, \quad (15)$$

and the convergence is exponentially fast [13]. The function V is called the *potential function* of q .

The evolution of ρ_t by FPE of the diffusion process can be interpreted as a continuous-time gradient flow under the $\mathcal{W}_2$ -metric in the probability space $\mathcal{P}_2$. The JKO scheme [39] computes a Wasserstein proximal GD which is a time discretization of the gradient flow. Specifically, define $G : \mathcal{P}_2^r \rightarrow \mathbb{R}$ as the KL divergence w.r.t. q , i.e.,

$$\begin{aligned} G(\rho) &= \text{KL}(\rho||q) = \mathcal{H}(\rho) + \mathcal{E}(\rho), \\ \mathcal{H}(\rho) &= \int \rho \log \rho, \quad \mathcal{E}(\rho) = c + \int V \rho, \end{aligned} \quad (16)$$

where c is a constant. More general G can be considered, see Section 4.1, and in this work we mainly focus on the case where G is the KL divergence as being considered in [39].

Under certain regularity condition of V , the JKO scheme computes a sequence of distributions ρ_n , $n = 0, 1, \dots$, starting from $\rho_0 \in \mathcal{P}_2$. For a fixed step size $\gamma > 0$, and the scheme at the n -th step can be written as

$$\rho_{n+1} = \arg \min_{\rho \in \mathcal{P}_2} G(\rho) + \frac{1}{2\gamma} \mathcal{W}_2^2(\rho_n, \rho). \quad (17)$$

The scheme computes the $\mathcal{W}_2$ -proximal Gradient Descent (GD) of G with step size γ , and can be written as

$$\rho_{n+1} = \text{Prox}_{\gamma G}(\rho_n). \quad (18)$$

The original JKO paper [39] proved the convergence of the discrete-time solution $\{\rho_n\}$ (after interpolation over time) to the continuous-time solution ρ_t of the FPE (8) when step size $\gamma \rightarrow 0+$. In the context of flow-based generative models by neural networks, the discrete-time JKO scheme with finite γ was adopted and implemented as a flow network in [74]. Our analysis in this work will prove the exponential convergence of ρ_n to q by the JKO scheme (including learning error), echoing the exponential convergence of the continuous-time dynamic (the FPE). This result leads to the guarantee of generating data distributions up to (TV) error $O(\varepsilon)$ in $O(\log(1/\varepsilon))$ JKO steps. We will summarize the flow model and introduce theoretical assumptions in Section 3.

3 Setup of JKO flow model and assumptions

In this section, we summarize the mathematical setup for the JKO flow model and introduce the necessary theoretical assumptions for our analysis. The guarantee of generating the data distribution will be derived in Section 5 based on the exponential convergence of the $\mathcal{W}_2$ -proximal GD (JKO scheme) in Section 4.

3.1 Forward and reverse processes of JKO flow model

As has been introduced in Section 1.1, the flow model implements an ODE model (transport equation), where both the forward process and the reverse process are computed by an invertible Residual Network [9] or a neural-ODE network [15, 29]. The forward process consists of N steps, where each step is computed by a Residual Block — in the neural-ODE model, this is the neural ODE integration on a sub-time-interval $[t_n, t_{n+1}]$, and we also call it a Residual Block. The backward process consists of the N steps of the same flow network “backward in time,” where each step computes the inverse map of the Residual Block, and in the neural-ODE model, this is via integrating the ODE in reverse time.

The forward and reverse processes (without inversion error) are induced by a sequence of transport maps, T_n , $n = 1, \dots, N$, which we will define more formally later. The two processes are summarized in (19),

$$\begin{aligned} \text{(forward)} \quad p &= p_0 \xrightarrow{T_1} p_1 \xrightarrow{T_2} \dots \xrightarrow{T_N} p_N \approx q, \\ \text{(reverse)} \quad p &\approx q_0 \xleftarrow{T_1^{-1}} q_1 \xleftarrow{T_2^{-1}} \dots \xleftarrow{T_N^{-1}} q_N = q. \end{aligned} \tag{19}$$

where p is the density of data distribution (when exists, otherwise a smoothified density by a short time diffusion), and q is the equilibrium density, typically chosen as Gaussian. Inversion error in the reverse process is considered in Section 3.3.

Forward process. In the forward process, the algorithm learns a sequence of T_n , which transports from data distribution P to the equilibrium distribution Q , which is typically the normal distribution, $\mathcal{N}(0, I)$. We denote by q the density of Q , and p the density of the data distribution P when there is one.

Following the neural-ODE framework used in [74], each step computes a transport map $T_{n+1} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ which is the solution map of the ODE from time t_n to t_{n+1} , i.e.,

$$T_{n+1}(x_n) = x(t_{n+1}), \tag{20}$$

where $x(t)$ solves $\dot{x}(t) = \hat{v}(x(t), t)$ on $[t_n, t_{n+1}]$, $x(t_n) = x_n$, and $\hat{v}(x, t)$ is the velocity field on $\mathbb{R}^d$ parametrized by the n -th Residual Block. Equivalently, we have

$$T_{n+1}(x_n) = x_n + \int_{t_n}^{t_{n+1}} \hat{v}(x(t), t) dt, \quad x(t_n) = x_n. \quad (21)$$

In the implementation of the JKO scheme in a flow network, the learning of the N Residual Blocks is conducted progressively for $n = 1, \dots, N$ by minimizing a training objective per step [74]. We emphasize that, unlike other normalizing flow models, which are trained end-to-end, the training procedure here is done step-wise and progressively over the N Residual Blocks.

Once T_{n+1} is learned, it pushes from p_n to p_{n+1} , i.e.,

$$p_{n+1} = (T_{n+1})_{\#} p_n. \quad (22)$$

In our problem, we want the distributions in the intermediate steps to have a density. To guarantee that this is the case for p_{n+1} , we technically need T_{n+1} to be *non-degenerate*. Intuitively, a non-degenerate map cannot collapse a set of finite (Lebesgue) measures into a set of measure zero.

Definition 3.1 (Non-degenerate map). *Denote by Leb the Lebesgue measure. $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is non-degenerate if for any set $A \subset \mathbb{R}^d$ s.t. $\text{Leb}(A) = 0$, then $\text{Leb}(T^{-1}(A)) = 0$.*

If a transport map is non-degenerate, then it pushes forward a density to a distribution that also has density, as shown in the following lemma proved in Appendix A.

Lemma 3.2. *Suppose $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is non-degenerate, $P \ll \text{Leb}$, then $T_{\#} P \ll \text{Leb}$.*

Assuming T_n are all non-degenerate, then the sequence of p_n produced by (22) all have densities, starting from $p_0 = p$ the data density. When data distribution has no density, we will introduce an initial short-time diffusion that mollifies the data distribution into ρ_δ which we set to be p_0 (see more in Section 5.1.2). The learning aims that after N steps, the final p_N is close to the equilibrium density q .

Reverse process (without inversion error). The reverse process computes the inverse of the N -steps transport by inverting each T_n in the forward process. We first assume that T_n can be exactly inverted in computation, which allows for a simplified analysis. In practice, T_n^{-1} can be implemented by fixed-point iteration [9] or reverse-time ODE integration [29]. The case when the inverse cannot be exactly computed is discussed in Section 3.3, where we need additional assumptions on the closeness of the computed inverse to the true inverse of T_n for our analysis.

The reverse process outputs generated samples, which are aimed to be close in distribution to the data samples, by drawing samples from q and pushing them through the reverse N steps. In terms of the sequence of probability densities generated by the process, the reverse process computes

$$q_n = (T_{n+1}^{-1})_{\#} q_{n+1}, \quad (23)$$

starting from $q_N = q$ and the output density is q_0 . Theoretically, the data processing inequality for the KL divergence applied to invertible transforms (Lemma 5.1) guarantees that if p_N is close to $q_N = q$, then q_0 is close to p_0 , which is the data density (possibly after short-time smoothing). This allows us to prove the guarantee of $q_0 \approx p_0$ once we can prove that of $p_N \approx q$, the latter following the convergence of the $\mathcal{W}_2$ -proximal GD up to a hopefully small learning error to be detailed below.